

ENTRY 035 Replication: Is It the Qubits, or the Chip? — August 16, 2026 — Status: COMPLETE

Entry 035 — A Second, Independent Batch Rules Out "Bad Qubits"

Two Experiments, Zero Shared Outlier Qubits, Same Pattern — This Is a Chip-Level Effect, Not a Qubit-Level One

1. The Question This Entry Answers

Entry 034's batch found 7 short routes (3-9 hops) that collapse completely to the 50% floor, clustered on Sherbrooke qubits 5 and 6. Before spending an entry chasing what makes those two qubits special, the proposed test was: run a second, independently-sampled batch with a different random seed and see whether the SAME qubits show up again (a qubit-specific defect worth isolating on its own) or whether a similar pattern reappears on DIFFERENT qubits (a general, chip-level effect worth chasing as one shared problem). entry035_replication_batch.py re-runs Entry 034's exact methodology with seed=99 instead of seed=7, drawing a fresh, non-overlapping set of qubit pairs.

2. Result: The Pattern Recurs, the Qubits Don't

	circuits	floor-collapse outliers	outlier qubits
Experiment 1 (Entry 034, seed=7)	116	7	97,114,95,72,3,6,2,5,28,86,91,11,22
Experiment 2 (Entry 035, seed=99)	122	5	83,103,82,84,73,80,40,56,15,9

Zero overlap. Not one physical qubit that produced a floor-collapse outlier in experiment 1 appears in experiment 2's outlier list, or vice versa -- despite both experiments sampling the same two chips with the same stratified hop-distance methodology. Overall gap statistics also matched closely between the two independent draws (mean 6.48 vs 6.10 points, gap-vs-distance correlation 0.20 vs 0.26), confirming this isn't two unrelated experiments -- it's the same underlying phenomenon sampled twice.

3. What Actually Distinguishes the Outliers: the Chip, Not the Qubit

Pooling both experiments and normalizing by how many short-route (≤ 10 hop) circuits were sampled on each chip:

chip	short-route circuits	floor-collapse	rate
Kyiv	65	2	3.1%
Sherbrooke	69	10	14.5%

Sherbrooke collapses to the floor on short routes roughly 4.7x more often than Kyiv, consistently across both independent samples. Both chips share the same uniform ecr-only basis and the same v4.1 model -- the difference has to come from something about Sherbrooke's actual topology or calibration surface that this project hasn't identified yet, not from a handful of bad qubits.

Entry 035 — Conclusion

The replication test gives a clear, decisive answer: floor-collapse is a general, recurring phenomenon rather than a defect isolated to qubits 5 and 6. It shows up on a completely different set of physical qubits in an independent sample, at a comparable rate, heavily skewed toward Sherbrooke over Kyiv. That rules out a qubit-specific patch as the right fix -- the next entry should isolate a FRESH instance of the collapse (not qubits 5/6, since those are now known not to be special) using the same gate-error / decoherence / readout separation Entries 028, 029, and 033 used, with the explicit goal of finding what property of the ROUTE or the CHIP -- not the qubit -- predicts it.

4. Next Step

Entry 036: pick 2-3 fresh floor-collapse instances from the combined pool of both experiments' outliers, isolate gate error, decoherence, and readout separately against Aer for each (as in Entries 028/029/033), and check whether the collapse traces to one specific channel consistently across instances despite the different qubits involved -- that would identify what the qubits have in common functionally, even without sharing physical identity.

Research Log — Index Addendum (Entry 035)

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Entry 035	Replication Batch — Floor-Collapse Is Chip-Level, Not Qubit-Level	Aug 16, 2026	Complete

Script: entry035_replication_batch.py (122 circuits, seed=99, reuses entry034_batch_dataset.py's methodology). Data: quantumbridge_data/entry035_replication_dataset.json, quantumbridge_data/entry035_comparison_summary.json. Zero qubit overlap with Entry 034's outliers; Sherbrooke's short-route floor-collapse rate (14.5%) is 4.7x Kyiv's (3.1%). Rules out a qubit-specific fix; next entry isolates channels on fresh instances.